

FUZHOUNESE DICTIONARY

A community dictionary of 福州話 · Foochow · Eastern Min

STATUS: **SHIPPED** · **LIVE AT FUZHOUNESE.ORG** · VERSION: **1.1** · AUTHOR: **NOAH LEE** · UPDATED: 2026-08-27

HOW TO USE THIS DOCUMENT

This is a *right-sized* PRD for a real community platform — a web frontend, a database, sign-in, file storage, and a moderation step. v1 is a single coherent release; every requirement is numbered so it can be pointed at (“implement FR-11”), everything speculative is a non-goal or an open question, and the intended reader is a developer new to the project. *v1 has now shipped and is live*; status chips mark what is built (**SHIPPED**), what was added after launch (**NEW V1.1**), and what remains undecided (**OPEN**).

WHAT CHANGED SINCE V1

- ▶ **Shipped & deployed.** The full v1 (Sections 1–7) is live at **fuzhounese.org**, on GitHub + Vercel with continuous deploys.
- ▶ **Words Wanted (v1.1).** A request-a-word / request-a-voice board with community upvotes, plus a homepage “words · voices recorded · words wanted” counter. See FR-23–FR-27.
- ▶ **About page (v1.1).** The project’s story and the maintainer’s bio. See FR-28.
- ▶ **Brand redesign (v1.1).** A letterpress visual identity — bone paper, lacquer-vermilion accent, grotesque display type over a serif body. See Section 6.

1. OVERVIEW **SHIPPED**

The Fuzhounese Dictionary is a web app where anyone can look up a Fuzhounese word — by Chinese characters, by romanization, or by its English or Chinese meaning — and where signed-in contributors can add new words. Every submission goes into a moderation queue and appears publicly only after an editor approves it. The dictionary data is openly licensed so the community can trust and reuse it. It is **live at fuzhounese.org**.

Problem. Fuzhounese (福州話 · Foochow · Eastern Min) has millions of speakers and a large overseas diaspora, but it is thinly documented online. What exists is scattered, inconsistent in romanization, and almost never lets ordinary speakers add the words they know. A heritage speaker who wants to check a spelling, find the characters for a word, or record a word before it is forgotten has nowhere central to go — and no way to contribute what they know back.

Who it’s for. Two primary audiences: **diaspora heritage speakers** who speak some Fuzhounese and want to look words up, confirm spellings/characters, and preserve vocabulary; and **language learners** who need pronunciations, meanings, and example usage. Linguists and researchers benefit but are not the group the product is designed around.

2. GOALS

SHIPPED

- G1** – A visitor can find a word by characters, romanization, or English/Chinese meaning in a few seconds, with tone marks optional.
- G2** – A signed-in contributor can add a fully-formed word entry in a couple of minutes, and it reaches an editor for review.
- G3** – An editor can approve or reject a submission in one action, and approved words appear on the site immediately.
- G4** – An entry can carry everything a real dictionary needs: characters, free-form romanization, IPA, audio, and one or more senses (each with an English definition and an optional Chinese gloss), plus examples and notes.
- G5** – Contributors are credited, and the dictionary's content is released under an open license (CC BY-SA).
- G6** – The whole thing runs on the free tiers of Supabase and Vercel.

3. NON-GOALS

Stating what we are *not* building is the highest-leverage part of this PRD. These fence off the features most likely to balloon the build.

- NG1** – No enforced romanization standard. Contributors write romanization however they know it; the app does not validate, convert, or auto-tone *Bàng-uâ-cê*. (Search is tone-insensitive, so this still works – see FR-2.)
- NG2** – No public corrections / edit-suggestion workflow in v1. Fixing an existing entry is an editor action; users cannot yet propose an edit to a specific word. **STILL OPEN** – revisit as traffic grows; Words Wanted is a first step toward structured community input.
- NG3** – No anonymous submissions. Contributing requires signing in.
- NG4** – No native mobile app. Responsive web is the platform.
- NG5** – No discussion, comments, forums, or social feed. Note: v1.1 adds *upvoting on word requests* (not on entries) – a narrow, moderation-friendly exception; see FR-25.
- NG6** – No full-sentence or machine translation. This is a word dictionary.
- NG7** – No offline mode.
- NG8** – No multi-tier editorial roles or permissions in v1. There is one owner-editor (see Q-4).

4. USER STORIES

SHIPPED

VISITOR (NO ACCOUNT)

- US-1** – As a visitor, I can search a word by Chinese characters, romanization, or English/Chinese meaning, so I can find it however I happen to know it.
- US-2** – As a visitor, I can open a word and see its characters, pronunciations (romanization, IPA, audio), meanings, and examples, so I understand it and can say it.

US-3 – As a visitor, I can browse all words alphabetically, so I can explore without a specific query.

US-4 – As a visitor, when my search finds nothing, I’m invited to contribute that word, so gaps get filled.

CONTRIBUTOR (SIGNED IN)

US-5 – As a visitor, I can sign in with Google, so I can contribute.

US-6 – As a contributor, I can submit a new word with all its fields, so I can grow the dictionary.

US-7 – As a contributor, I can attach an audio pronunciation (upload a file or paste a link), so others can hear the word.

US-8 – As a contributor, I can add more than one sense to a word, so a word with several meanings is captured fully.

US-9 – As a contributor, I can see which of my submissions are pending, approved, or rejected, so I know the status of my work.

US-10 – As a contributor, I’m credited on entries I add, so my contribution is visible.

EDITOR

US-11 – As an editor, I can see all pending submissions in one queue, so I can review them.

US-12 – As an editor, I can approve or reject a submission with an optional note, so quality stays high and the contributor learns why.

US-13 – As an editor, I can edit any entry, so I can fix mistakes without a separate corrections system.

US-14 – As an editor, I can bulk-import my own pre-approved word list, so I can seed a few hundred words quickly.

5. FUNCTIONAL REQUIREMENTS

SHIPPED

Numbered, explicit, one behavior each. If a requirement needs the word “and” twice, it’s split.

SEARCH & DISPLAY

FR-1 – The home page has a single search box. Submitting a query returns approved entries whose characters, romanization, IPA, English definition, or Chinese gloss match the query, ranked by relevance.

FR-2 – Search ignores tone diacritics and letter case: `sioh` matches `siōh`, and `hok` matches `Hók`.

FR-3 – Each search result shows the characters (if any), the romanization, the part of speech, and the first sense’s short gloss, and links to the full entry.

FR-4 – An entry page shows: characters, romanization(s), IPA, an audio player (if audio exists), and each sense with its part of speech, English definition, Chinese gloss, example sentence, example translation, and notes.

FR-5 – When an entry has no Chinese characters (many colloquial Min morphemes have no standard hàn-cê), the character field is simply omitted; the entry is still valid and fully displayable from its romanization alone.

FR-6 – A browse page lists all approved entries alphabetically by romanization, paginated.

FR-7 – When a search returns zero results, the page shows a “Know this word? Add it” call to action that pre-fills the submission form with the query. If the visitor is not signed in, it prompts sign-in first.

ACCOUNTS & SUBMISSION

FR-8 – A user can sign in with Google (OAuth, via Supabase Auth). There are no passwords.

FR-9 – Only signed-in users can open or submit the “Add a word” form. Signed-out users see a sign-in prompt in its place.

FR-10 – The submission form accepts, per entry: characters (optional), romanization (optional, free-form), IPA (optional), and audio (optional – a file upload or a URL); plus one or more *senses*. Each sense has: part of speech (optional), English definition (**required**), Chinese gloss (optional), example sentence (optional), example translation (optional), and notes (optional).

FR-11 – A submission is valid only if it has at least one identifying form (characters or romanization) *and* at least one sense with an English definition. An invalid submission shows an inline error and saves nothing.

FR-12 – Audio upload accepts common audio formats up to a fixed size limit (target 5 MB); a pasted audio URL is stored as-is. Uploaded files go to a Supabase Storage bucket.

FR-13 – On submit, the entry is saved with status `pending` and stamped with the contributor’s user id, and the contributor sees a confirmation.

FR-14 – A contributor has a “My submissions” view listing their entries with each one’s status (pending / approved / rejected) and any editor note.

MODERATION

FR-15 – Editors reach a moderation queue at a protected route, listing pending submissions oldest-first with every field shown and any attached audio playable inline.

FR-16 – An editor can **approve** a submission (status → `approved` ; it appears publicly at once) or **reject** it (status → `rejected` ; hidden but kept on record), with an optional review note shown to the contributor.

FR-17 – An editor can edit any entry’s fields at any time, whatever its status.

FR-18 – An approved entry shows a contributor credit (display name), except for entries the owner bulk-imported.

FR-19 – Editors can bulk-import entries from a CSV as status `approved` , bypassing the queue.

CONTENT & PLATFORM

FR-20 – Every page footer states the content license (CC BY-SA) and links to it.

FR-21 – The layout is usable at 375px wide (phone) and 1280px wide (laptop) without horizontal scrolling.

FR-22 – Row-Level Security guarantees the public/anonymous role can read only `approved` entries – never pending or rejected ones – and that approve, reject, and edit happen only through a privileged server path.

COMMUNITY – WORDS WANTED

NEW V1.1

FR-23 – A public `/wanted` board lists words the community wants added or recorded, most-requested first. Two kinds of request: a **new word** that isn’t in the dictionary yet, and a **pronunciation** for an existing entry that has no audio.

FR-24 – Any signed-in user can create a request (a word/phrase plus an optional note). An entry page with no audio shows a “Request a pronunciation” action that files a request tied to that entry.

FR-25 – Signed-in users can upvote a request; each user counts once. Duplicate requests are de-duplicated – asking for a word that already has an open request adds a vote instead of a second row. (Narrow exception to NG5; votes are on requests, never on dictionary content.)

FR-26 – An editor can mark a request **fulfilled**. Requests carry no personal data beyond the requester’s id and are governed by the same Row-Level Security model as the rest of the app.

FR-27 – The home page shows a live tally: total approved words, entries with a recording (“voices recorded”), and open requests, with the last linking to `/wanted`. The tally degrades gracefully if the requests table isn’t present.

ABOUT & IDENTITY

NEW V1.1

FR-28 – An `/about` page tells the project’s story and shows the maintainer’s name, photo, and a short bio. It is reachable from the main navigation.

FR-29 – The site presents a consistent wordmark – 福州話 **fuzhounese.org** – in the header on every page, and a browser-tab title.

6. DESIGN CONSIDERATIONS

REDESIGNED V1.1

v1.1 replaced the plain default styling with a deliberate **letterpress identity** drawn from Chinese lexicographic and print tradition. The reading-first principles below are unchanged; the visual language is new.

BRAND IDENTITY (V1.1)

- **Palette.** A cool bone-paper ground, near-black ink, and a single **lacquer-vermilion** accent (the red of Fuzhou lacquerware) for headwords, links, and calls to action. Semantic colors stay separate from the accent.
- **Type.** A tight, uppercase **grotesque** display face (Bricolage Grotesque, standing in for Dada Grotesk) for headlines and headwords; a **serif** (Newsreader) for body and definitions; a **monospace** (IBM Plex Mono) for labels, eyebrows, and IDs.
- **Structure.** Hard edges and hairline rules instead of rounded cards; each dictionary entry is set like a real dictionary entry – headword in the display face, romanization in lacquer, part of speech in an italic tag.
- **Themes.** Full light and dark palettes via CSS-variable tokens, following the OS preference.

READING PRINCIPLES (UNCHANGED)

- Reading-first and quiet. A single column on reading pages; search results may run wider. The visual hierarchy inside an entry is always character → pronunciation → meaning.
- Characters are shown large in the display/CJK face; romanization in the accent color; meanings in plain body text.
- Tone diacritics must render correctly, including stacked combining marks (ǖ, è, ā, ě). Use a Unicode-complete font stack and verify these glyphs on a real device.
- Missing data degrades gracefully: absent characters, audio, examples, or glosses simply disappear rather than leaving empty labels.
- Respect the OS light/dark preference. Audio is a plain native player. On mobile, the search box is the hero element.

7. TECHNICAL CONSIDERATIONS

SHIPPED

TC-1 – Next.js (App Router, TypeScript) frontend; Supabase (hosted Postgres + Auth + Storage) backend; deployed on Vercel. Runs on the free tier of each.

TC-2 – Senses are a child table, not a JSON blob, so search and display treat every sense uniformly and a word can have many. Data shape (simplified):

```
entries { id, headword, hanzi?, romanization?, ipa?, audio_url?,
          notes?, variety?, status, contributor_id, created_at, reviewed_at }
senses  { id, entry_id, part_of_speech?, definition_en,
          gloss_zh?, example?, example_gloss?, sort }
profiles { id (= auth uid), display_name, is_editor }
```

TC-3 – Search is a Postgres function combining `unaccent` + trigram (`pg_trgm`) matching across entry fields and senses, so both tone-insensitive and partial matches work; the client calls it via RPC.

TC-4 – Auth is Supabase Auth with Google OAuth. A `profiles` row is created on first sign-in. An `is_editor` boolean, set manually, unlocks the moderation queue and edit actions; the column is revoked from client roles so no one can self-promote.

TC-5 – Row-Level Security: anyone may `SELECT` only `status = 'approved'`; an authenticated user may `INSERT` an entry as `pending` with `contributor_id` equal to their own uid; approve / reject / edit are restricted to the editor role (or performed through a service-role server action).

TC-6 – Audio lives in a Supabase Storage bucket: authenticated users may upload to their own folder; files are publicly readable; the entry stores the public URL. Size and type are enforced on the client and by a bucket policy.

TC-7 – No secret keys ever reach the browser. The service-role key is used only in server code paths.

TC-8 **NEW V1.1** – **Deployment.** Source is a public GitHub repo (`fuzhounesedictionary`); Vercel is connected for continuous deploys on every push to `main`. Custom domain **fuzhounese.org** (apex A record + `www` CNAME to Vercel), auto-SSL. Env vars in Vercel: the two public Supabase values, the service-role key, and `NEXT_PUBLIC_SITE_URL`. Supabase Auth Site URL / Redirect URLs include the production domain and localhost. Database schema changes are applied manually in Supabase (not part of the deploy).

TC-9 **NEW V1.1** – **Words Wanted schema.** A `word_requests` table (term, optional `entry_id`, note, status open|fulfilled, requester) and a `word_request_votes` table (one row per user per request). Partial-unique indexes keep at most one open request per entry and per free-text term. A `security_invoker` view joins vote counts and the linked entry's current audio state for ranking.

TC-10 **NEW V1.1** – **Fonts.** Bricolage Grotesque, Newsreader, and IBM Plex Mono are self-hosted via `next/font` (fetched at build time), exposed as CSS variables and mapped in Tailwind.

8. SUCCESS METRICS

SM-1 (primary – coverage) – The dictionary grows from its initial seed (a few hundred words) toward 1,000+ approved entries, and the live count is visible on the home page.

SM-2 (primary – findability) – For a fixed set of words known to be in the dictionary, searching *any* of their forms – characters, romanization with or without tone marks, or English – returns the entry in the top results.

SM-3 (secondary – community) – A measurable share of newly approved words come from community submissions, not only the owner’s bulk import.

SM-4 (secondary – usage) – Visitors return and look words up: a basic count of searches per week and entry views trends upward after launch.

SM-5 NEW V1.1 (secondary – participation) – Words Wanted is used: open requests are created and upvoted, and fulfilled requests convert into new entries or recordings over time.

9. OPEN QUESTIONS

Q-1 – Which variety is the reference? “Fuzhounese” spans the city dialect and county varieties (Changle, Lianjiang, Fuqing/Hokchia, Gutian, Matsuo...). *Current approach: accept all, with an optional “variety / region” tag on each entry. Whether to canonicalize the Fuzhou-city form as the headword is still open.* OPEN

Q-2 – Duplicates. With no corrections flow (NG2), what happens when a contributor submits a word that already exists – reject, allow a second entry, or route to the editor to merge? *Assume: the editor decides at review; a “possible duplicate” hint in the queue is a candidate improvement.* OPEN

Q-3 – Chinese gloss. Is it specifically Mandarin, and is it ever required? *Assume: an optional Mandarin gloss.* OPEN

Q-4 – Editorial team. v1 assumes a single owner-editor. When do we need multiple editors with roles, and an audit of who approved what? *Out of scope for v1; flag when a second reviewer is needed.* OPEN

Q-5 – Spam & abuse. Beyond requiring sign-in, do we need rate limits, a captcha, or a trusted-contributor tier as volume grows? *Assume: sign-in + moderation are enough for now. A quality-weighted contributor reputation is a candidate.* OPEN

Q-6 – Name & domain. *Resolved: the site is fuzhounese.org, with the wordmark 福州話 fuzhounese.org.* RESOLVED V1.1

Q-7 – Audio review load. Audio makes moderation heavier – the editor has to listen. Is audio reviewed inline in v1 (current assumption), or added only by editors if community audio proves noisy? *Words Wanted’s request-a-voice flow concentrates demand where it’s wanted; a community listen-and-verify queue is a candidate.* OPEN

NOTES ON WHY THIS DOCUMENT IS SHAPED THE WAY IT IS

It is scoped to one buildable release. This is a platform – frontend, database, auth, storage, and a moderation step. The response was not a bigger, vaguer document but a tighter one: v1 is a single coherent release, and everything past it lives in Non-goals or Open questions. That v1 has now shipped, and the additions since (Words Wanted, About, the redesign) slot into the same numbered structure rather than sprawling.

Non-goals do the heavy lifting. With any capable builder, the failure mode is too much, not too little – a corrections workflow, multi-editor roles, and romanization validation are exactly the features that would triple the build. Section 3 is the fence, and it held: the one deliberate crossing (upvotes, FR-25) is narrow and stated.

Every requirement is numbered and testable. “Search should be forgiving” is not a requirement; FR-2 is, because you can write down an input (sioh) and its expected output (siöhh).

Ambiguity is named, not hidden. Section 9 says out loud what hasn’t been decided — above all, which variety of Fuzhounese is the reference. A builder who hits an undecided question will decide it silently; this document hands those decisions back to you, and records the ones now settled.